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People systematically under- and overestimate public engagement in climate action

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Globally, people underestimate others' support for climate action—a phenomenon frequently described as pluralistic ignorance, which is seen as a major barrier to collective climate engagement. However, little is known about whether pluralistic ignorance occurs regarding actual behaviour, what mechanisms underlie it and whether correcting it changes such behaviours. Here, across five preregistered experiments ($N = 5{,}081$), we found that participants overestimated rare actual behaviours such as others' donating or political engagement for climate action, while we replicated the underestimation of broad public support. Misestimations arise primarily from general cognitive processes when estimating proportions. Finally, correcting these misperceptions partly affected self-reported donation willingness but not actual donations. These findings suggest that pluralistic ignorance in the climate domain may be more intricate and less consequential than previously thought.

As humans cause climate change, it is pivotal to understand and promote human attitudes and behaviours to mitigate it. Climate action takes many forms, ranging from individual lifestyle changes to supporting climate policies and engaging in political action[1]. In line with theories, psychological literature has shown that social norms (that is, perceptions of others' attitudes and behaviours) can strongly influence attitudes and behaviours[2–6]. Consequently, social norms have been identified as a major force for understanding and promoting climate action[6–8].

Research on perceptions of climate-related attitudes and behaviours has converged on one finding: the vast majority of people around the world support climate action, but people considerably underestimate the extent of this support[9–14]. This phenomenon, frequently labelled pluralistic ignorance, has attracted increasing attention in the climate field (Supplementary Section 1) for two reasons. First, it may intimidate people from talking about climate change, thereby undermining climate action[15,16]. Second, correcting misperceptions through communicating actual support has been proposed as a cost-effective way to promote climate action[9,10,13]. However, research on pluralistic ignorance has primarily focused on self-report measures, which may misrepresent actual climate-action behaviour. Here, we therefore provide a more comprehensive understanding of the phenomenon by studying its generalizability to actual behaviour—the most relevant expression of climate action—and its mechanisms and consequences.

Studies have shown that people worldwide consistently underestimate the large share of people who believe in climate change[11,17], worry about it[13], want stronger government engagement[18], support climate policies[13], report climate-friendly behaviour[9] and are even willing to contribute 1% of their household income every month to fight climate change[10]. These findings suggest that perceptions of public climate action are generally biased towards underestimation, leading to the conclusion that "the world is in a state of pluralistic ignorance"[10]. However, these studies rely predominantly on self-report measures and neglect the actual, impactful behaviours needed to effectively reduce carbon emissions (for example, political engagement). As people's willingness, intentions and attitudes do not directly translate into actual behaviour[19,20], our first goal was to examine whether pluralistic ignorance generalizes to actual behaviour required for societal transformation.

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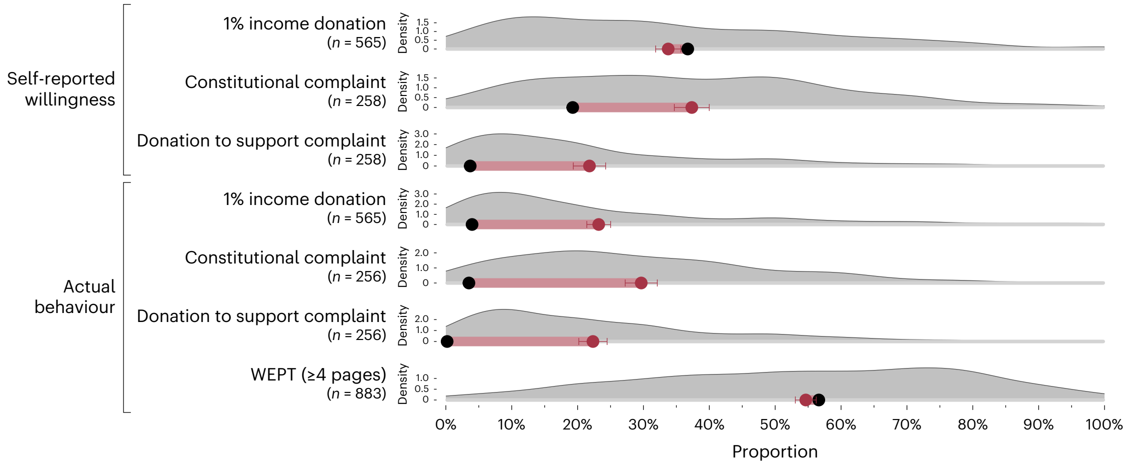

Fig. 1 | Pluralistic ignorance in climate willingness and behaviours (Studies 1 and 2). People overestimate rare climate-action behaviour but underestimate more common willingness and behaviours. Black dots represent actual proportions; red dots represent estimated means. Error bars represent 95% confidence intervals. The red line represents the pluralistic-ignorance gap. The grey area represents the distributions of estimations. For each outcome, group sizes (n) represent the number of participants providing an estimate. WEPT, Work for Environmental Protection Task[34,35].

The consistent pattern of underestimation is primarily attributed to a biased environment, including misrepresentative media coverage, lobbying and a lack of conversations about climate change[10,13,15,21], suggesting underrepresentation of climate action. This account, therefore, predicts that people generally underestimate climate action, regardless of how it is measured or how large the actual proportion of people holding certain views or performing behaviours is. Crucially, most studies examine attitudes and intentions of large majorities (that is, prevalences >60%). However, underestimating large population proportions is not unique to climate action: when people are asked to estimate large groups (for example, washing-machine owners in the USA), they generally underestimate them[22,23]. This pattern is thought to reflect rational cognitive processes: when estimating proportions under uncertainty, people not only consider available information and knowledge about the specific group but also incorporate previous expectations about typical group sizes. This expectation can refer to typical sizes of demographic groups but can also be more specific, such as when judging about groups related to climate action or known to be minorities[22]. Consequently, estimates are shifted towards the mean of these expectations[22–24], typically moving extreme values towards less extreme estimates. The extent to which estimates are shifted towards the mean is usually greater the less certain a person is about a proportion. This regression-to-the-mean account, too, would predict underestimation of large proportions of climate-action supporters. However, the regression-to-the-mean and the biased-environment accounts make opposite predictions for small proportions common in impactful climate action, such as the share of protestors or vegans[25–28]: whereas the biased-environment account predicts underestimation, the regression-to-the-mean account predicts overestimation. Therefore, our second goal was to test whether regression to the mean contributes to pluralistic ignorance.

If misperceptions cause people to refrain from climate action as commonly argued, correcting these misperceptions could be a promising way to foster climate action[10,15]. Yet, empirical evidence is weak regarding whether communicating prevalent support for climate action indeed increases climate-related attitudes and behavioural intentions (with very small[9,29], inconsistent[15,29] or null effects[17,30–32]), and studies on actual behaviours are very rare[9,30]. For this reason, the third goal of this research was to test if and how correcting misperceptions by communicating descriptive norms of climate action—either high willingness or low actual behavioural prevalence—affects impactful climate behaviour.

In sum, this research provides a more comprehensive understanding of pluralistic ignorance in climate action, its generalizability, mechanisms and consequences. Specifically, this research (1) examines whether pluralistic ignorance generalizes to actual behaviour, (2) tests whether regression to the mean contributes to pluralistic ignorance and (3) assesses how communicating high and low prevalences of climate action (descriptive norms) affects actual impactful climate behaviour.

Results

We conducted five preregistered experiments (total $N = 5{,}081$) to study pluralistic ignorance in climate action. Detailed results are presented in Supplementary Section 2. Preregistrations, materials, data and code can be found at https://osf.io/kcq37 (ref. 33).

Generalizability of pluralistic ignorance

To test whether pluralistic ignorance extends to actual behaviour, study 1 ($N = 1{,}130$) adapted a paradigm from Andre et al.[10]. A quota-representative German sample self-reported whether they would be willing to donate 1% of their net household income monthly to fight climate change. In addition to the original paradigm, willing participants could actually donate this amount immediately to a climate-action programme. Then, half of the participants estimated the share of others willing to donate and the other half estimated who actually did. Overall, 36.7% reported willingness to donate but participants estimated that only $M = 33.8\%$ of participants would, indicating slight underestimation (gap: −3.0 percentage points (p.p.), $P = 0.002$; Fig. 1). For actual behaviour, the pattern reversed: 3.7% verifiably donated, yet participants estimated that $M = 23.2\%$ would do so (+19.5 p.p., $P < 0.001$). Thus, willingness was slightly underestimated, whereas actual donations were strongly overestimated.

To replicate overestimation of actual behaviour in another domain, study 2 ($N = 628$) examined participation as a claimant in a constitutional complaint against the German government demanding stricter climate policies (https://zukunftsklage.greenpeace.de/). Participation in the complaint required no financial resources or substantial effort. Participants could participate in the complaint and/or donate to support it. Again, half estimated the share of others willing to participate; the other half estimated who actually did. Overall, 19.3% indicated willingness to join (an additional 2.5% reported having already joined), yet this was overestimated ($M = 37.3\%$, +18.0 p.p., $P < 0.001$; Fig. 1). Only 3.5% verifiably participated, but participants

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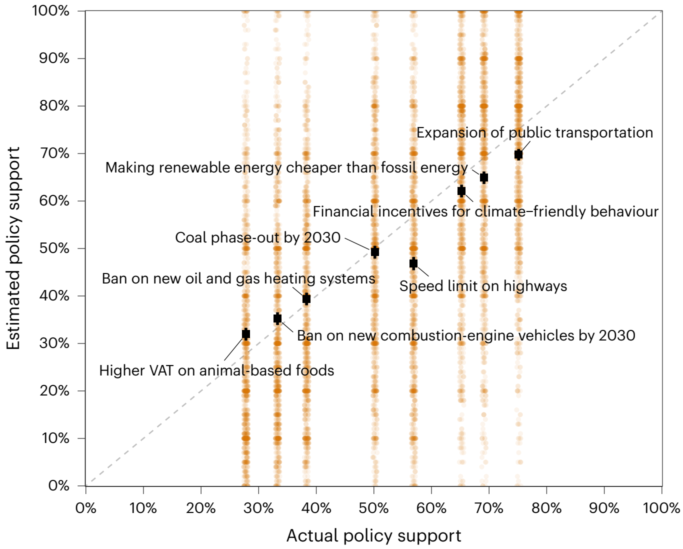

Fig. 2 | Over- and underestimation of policy support (study 1). People underestimate support for policies with high support and overestimate support for policies with low support ($N = 1{,}130$). Squares represent mean estimations, and error bars represent 95% confidence intervals. Points represent raw data, slightly jittered for illustration. Points below the diagonal represent underestimation and points above the diagonal represent overestimation. VAT, Value-Added Tax.

estimated $M = 29.7\%$ (+26.2 p.p., $P < 0.001$). Willingness to donate (3.7%; 3.3% reported having already donated) was likewise overestimated ($M = 21.8\%$, +18.1 p.p., $P < 0.001$), as were verified donations (0.2%, $M = 22.3\%$, +22.1 p.p., $P < 0.001$; Fig. 1). Thus, both political participation willingness and behaviour were consistently overestimated.

In sum, Studies 1 and 2 revealed overestimation of rare, high-impact behaviours but less consistent patterns for self-reported willingness. This could reflect general overestimation of actual behaviour—or, alternatively, overestimation is due to the proportions of actual behaviours being small, in line with the regression-to-the-mean account.

The role of regression to the mean

To distinguish between the two accounts—general overestimation of behaviour versus regression to the mean—we next examined actual behaviour performed by the majority in a follow-up of study 1 ($n = 883$). Participants completed the Work for Environmental Protection Task[33,34] (WEPT), in which they could invest effort in a number-identification task to generate donations to an environmental organization. The WEPT is a validated paradigm that captures actual pro-environmental behaviour[34–36]. Participants worked on up to six pages. The more pages they completed, the more donations they triggered. They then estimated the proportion of participants completing at least four pages—a behaviour expected of the majority[32,34,35]. The majority (56.6%) did so, yet participants did not overestimate this but tended to underestimate this ($M = 54.6\%$, −2.0 p.p., $P = 0.014$; Fig. 1). Thus, we found slight underestimation—rather than overestimation—of actual behaviour performed by a large proportion of people, supporting the regression-to-the-mean account.

To test whether regression to the mean extends beyond actual behaviour, study 1 also assessed support for eight climate policies with low, intermediate and high public support[37]. In addition, participants also estimated others' policy support. If regression to the mean also applied to attitudinal policy support, participants should over- and underestimate support for policies supported by a small and large proportion of people, respectively. Estimates regressed towards the mean ($P < 0.001$; Fig. 2 and Supplementary Section 2): Support for the two least popular policies was slightly overestimated ($P_s < 0.005$), while support for the four most popular policies was slightly underestimated ($P_s < 0.001$), with no bias for intermediate ones ($P_s > 0.121$). Thus, regression to the mean applied to both behaviours and attitudes.

To test the robustness of the regression-to-the-mean account, we tested it across a broader range of outcomes with high and low values (for example, popular versus unpopular policies and frequent versus rare behaviours). In study 3 ($N = 1{,}112$), a quota-representative US sample reported frequencies of 20 climate-friendly behaviours (for example, buying energy-efficient appliances), support for 20 policies (for example, a ban on combustion-engine cars), completed a ten-page WEPT and decided how much of a bonus (either US$1 for sure or US$50 to win, randomly assigned) to donate to a climate organization. Next, participants estimated the percentage of participants performing the climate-friendly behaviours, supporting the climate policies, either completing at least one WEPT page or all ten pages, and either donating some amount or the full amount, representing majority and minority behaviours, respectively.

We found a consistent pattern across all outcome measures: small proportions were overestimated and large ones were underestimated (Fig. 3 and Supplementary Section 2), indicating a regression-to-the-mean pattern. Exploratory analyses showed that both people who did and those who did not belong to the group being estimated overestimated very small proportions and underestimated very large ones, suggesting that this pattern was not driven by misestimations of a particular subgroup[21]. To corroborate the regression-to-the-mean account, we computationally modelled the deviations between estimated and actual proportions using a hierarchical extension of the Uncertainty-Based Rescaling model[22], which formalizes regression to the mean (Methods). This model described our data well (Bayesian $R^2 = 0.53$; Fig. 3), supporting that regression to the mean underlies misestimations of climate action.

To examine the interindividual and environmental factors shaping estimates, we explored associations of participants' individual model parameter estimates with individual characteristics (that is, demographics and individuals' climate action as measured in the study) and perceptions of discussions and media coverage about climate change. Larger values of the parameter δ (delta) reflect an upwards shift in people's estimates (that is, that a person thinks a larger share of people are climate active across the board or, in Bayesian terms, a higher mean prior expectation). δ was most strongly positively associated with one's own climate action (in terms of donations, $\beta = 0.28$, $P < 0.001$, and individual behaviour, $\beta = 0.13$, $P = 0.003$). Further, people experiencing more frequent climate change-related discussions ($\beta = 0.15$, $P < 0.001$) and media coverage ($\beta = 0.12$, $P = 0.002$) provided higher estimates. This suggests that people's own engagement[11,13,21] and experiences[13] influence perceptions beyond regression to the mean.

The parameter γ (gamma) indicates how well people differentiate between what is common versus rare (that is, the degree of certainty), which typically corresponds to greater estimation accuracy. Greater differentiation was observed among older ($\beta = 0.25$, $P < 0.001$) and more educated participants ($\beta = 0.18$, $P = 0.014$), those donating more ($\beta = 0.10$, $P = 0.002$) and those experiencing fewer climate discussions ($\beta = -0.23$, $P < 0.001$). This suggests that knowledge and experience may improve differentiation.

In sum, estimation biases across self-reported and actual behaviours and attitudes were well captured by regression to the mean and further shaped by one's own behaviour and perceived environment, with differentiation increasing with age and education. We next examined the behavioural implications of correcting misperceptions.

Communicating descriptive norms of climate action

To test how correcting overestimation of rare climate-action behaviour by communicating low descriptive norms affects climate action, in the study 1 follow-up, half of the participants were informed that 4% of participants in the initial survey had actually made a monthly donation of 1% of their household income, while the other half received no

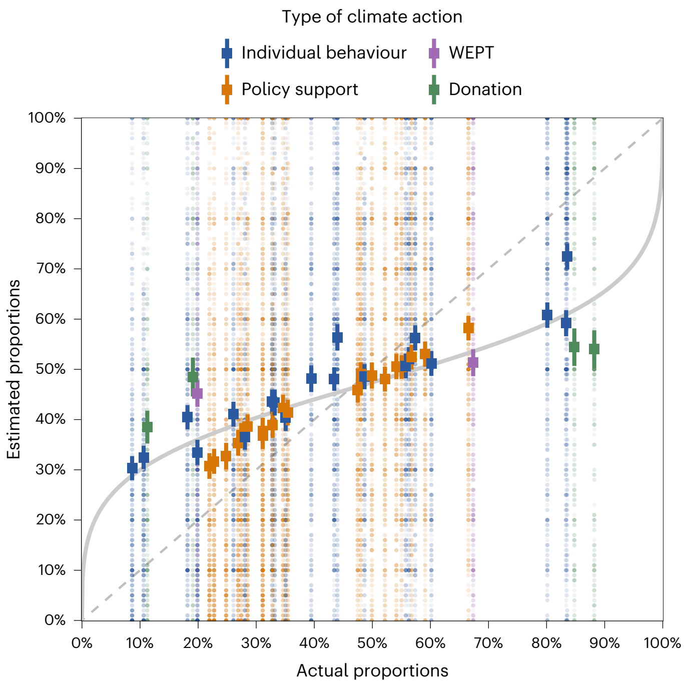

Fig. 3 | Over- and underestimation of climate action (study 3). Across types of climate action, participants underestimated large proportions and overestimated small proportions (n_s = 252–496). Squares represent mean estimates; error bars represent 95% confidence intervals. Points represent raw data. Points below the diagonal represent underestimation and points above the diagonal represent overestimation of the actual proportions. The solid curve represents predictions by the hierarchical variant of the Uncertainty-Based Rescaling model[22] based on the estimated mean group-level parameter values (δ = 0.86 (95% highest-density interval 0.78–0.94); γ = 0.34 (0.31–0.37)). The hierarchical version of the Uncertainty-Based Rescaling model described our data well (Bayesian R^2 = 0.53), indicating that the pattern of deviations between estimated and actual proportions can be well described by regression to the mean. This is true across outcomes. WEPT, Work for Environmental Protection Task[34,35].

information. Providing this low prevalence did not affect intentions to discuss climate change, sign petitions or donate (P_s > 0.091; Fig. 4a), suggesting that communicating low prevalence may not affect intentions to act.

To replicate and extend this finding with a high-prevalence comparison and test the effect on actual behaviour, we conducted studies 4a (N = 1,092) and 4b (N = 1,119). In both, German participants (quota-representative) received information that either 4% of participants actually made monthly donations of 1% of their household income (low prevalence; data from study 1), that 68% were willing to contribute that amount (high prevalence; data from Andre et al.[10]) or no information. We communicated those real empirical results to vary prevalences while avoiding deceiving participants. We then measured donation willingness and behaviour, as in study 1. Across studies, self-reported willingness to donate increased only by communicating high-prevalence willingness (P < 0.001; low prevalence: P = 0.476), but neither norm communication affected actual donations (P_s > 0.786; Fig. 4b,c). Thus, correcting misperceptions in either direction neither increased nor decreased actual climate action.

In sum, communicating broad support for climate action increased self-reported climate action but not actual behaviour. Communicating low engagement did not affect self-reported or actual climate action.

Discussion

Previous research has consistently found that people underestimate others' support for climate action, a phenomenon often labelled pluralistic ignorance. It has mainly been viewed as resulting from biased information environments and as a barrier to collective climate action, prompting calls to correct such misperceptions. This research provides a more comprehensive account of this phenomenon by examining its generalizability, mechanisms and consequences. Across five preregistered experiments, we found pluralistic ignorance happening both ways (that is, replicating the well-known underestimation effect in frequent attitudes and behaviours but also showing overestimation of rare ones). These distortions appear to arise primarily from general cognitive processes when estimating proportions. Finally, this research showed that correcting misperceptions did not affect impactful climate action.

Rigorous surveys have repeatedly shown that people underestimate widespread public support for climate action[9–15] but have largely neglected actual behaviour. In line with the well-known intention–behaviour gap[19,20], we found broad attitudinal support and willingness translated into specific action (for example, donating to a climate-action organization) only among a small minority—and this minority was consistently overestimated. In fact, overestimation of relatively rare climate-friendly behaviours and unpopular policies has been found before[9,38,39] but not identified as part of a consistent pattern. Thus, misperceptions do not uniformly take the form of underestimation but depend on the outcome in question. These findings complement previous evidence that qualitatively different patterns can occur when comparing intentions and behaviour[40,41] and underscore the need to study impactful, real-world behaviours rather than attitudes or intentions alone[19].

Our research suggests a unifying explanation for both over- and underestimation: When estimating proportions under uncertainty, people incorporate previous expectations about population sizes besides knowledge about the specific attitude or behaviour, resulting in extreme values being estimated as less extreme, a pattern also found in estimations in various other domains than climate[22,23]. In this account, misestimates therefore reflect rational cognitive processes rather than bias. Supporting this account, older and more educated participants provided more differentiated estimates, suggesting that greater knowledge reduces uncertainty in estimations. While this account could explain the qualitative pattern of our results, we found evidence for additional factors shaping estimations. Consistent with the prevailing explanation for pluralistic ignorance, the systematic underrepresentation of public support in the environment[10,13,15,21], participants who encountered climate change-related discussions or media coverage more frequently also estimated more people to be climate active. This is also illustrated by estimates of support for the widely debated speed limit on German highways noticeably deviating from the regression-to-the-mean pattern (Fig. 2). Although climate-related media coverage and conversations may overrepresent highly visible minorities such as protesters, the consistent pattern across diverse forms of climate action—including less publicly visible behaviours—suggests that media overrepresentation alone is unlikely to account for the observed overestimations. In line with previous research, we also found evidence for a false-consensus effect[11–13,21,42]: if people were more climate active themselves, they also believed more people to be climate active. This suggests that regression to the mean and other factors such as false consensus are complementary: estimates are generally pulled towards the mean of the previous expectation, but this expectation seems to depend on individual differences in climate-action support and environmental perceptions.

Communicating that the majority is willing to contribute financially to fighting climate change increased general willingness to donate but not actual donations to a specific climate-action organization. Communicating low prevalences of actual donation behaviour, on the contrary, did not affect behavioural intentions, willingness to donate or actual donation behaviour. A possible explanation for this discrepancy is that our tested behaviour is regarded as socially desirable, as it has been shown that communicating an injunctive norm (for example, that saving energy is desirable) can buffer the negative effect

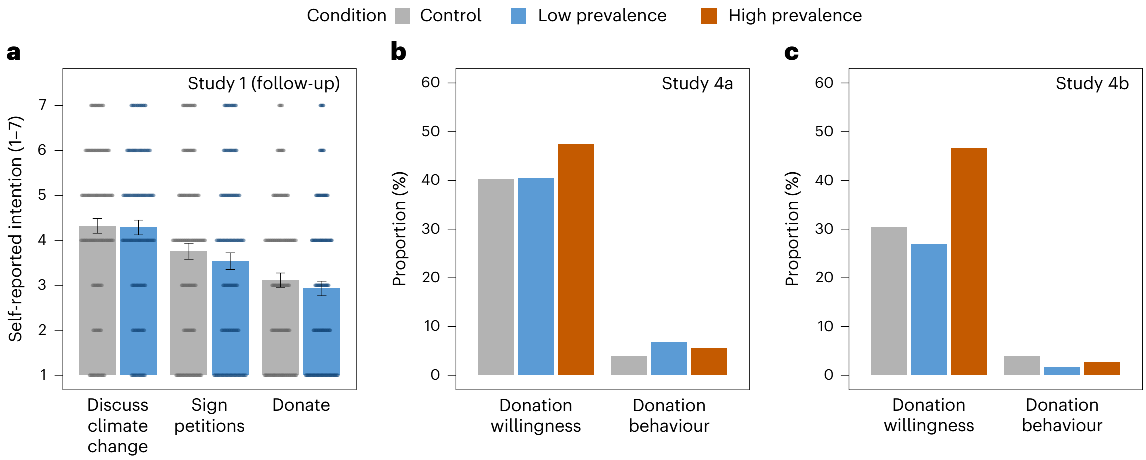

Fig. 4 | Effects of norm communication on climate action intentions, willingness and behaviour (studies 1, 4a and 4b). a, Communicating a low-prevalence descriptive norm of climate action did not affect intentions to discuss climate change, sign petitions or donate in the study 1 follow-up (control, $n = 442$; low prevalence, $n = 441$). Bars represent mean values, and error bars represent 95% confidence intervals. Points represent raw data. **b**,**c**, Communicating a high-prevalence descriptive norm increased donation willingness but not behaviour; communicating a low-prevalence norm did not affect donation willingness or behaviour in studies 4a (**b**) (control, $n = 365$; low prevalence, $n = 354$; high prevalence, $n = 373$) or 4b (**c**) (control, $n = 378$; low prevalence, $n = 362$; high prevalence, $n = 379$).

of low descriptive norms (for example, that others consume more energy than oneself)[43–46]. Furthermore, it is possible that highlighting the proportion of people not donating (that is, framing the descriptive norm differently) would have led to different results[47,48]. Taken together, our results suggest that a brief communication of norms may be too weak to change behaviour.

At the same time, this research reveals other, perhaps more promising, avenues for promoting climate action. While most people support climate action in principle and are willing to contribute to climate action, relatively few endorse stricter policies or invest substantial resources. This discrepancy offers considerable potential for behavioural interventions. Instead of focusing on the remaining small minority of climate sceptics, behavioural science should prioritize the much larger group of supporters not yet acting and seek to close the attitude–behaviour gap. Evidence-based strategies can help achieve this[49]: facilitating climate-friendly behaviour through convenience and subsidies[50,51]; increasing policy support through fair design[52–55], transparent explanations[56,57] and communicating effectiveness and cobenefits[58–63]; and encouraging donations by highlighting their impact[64]. Given widespread public support and a well-stocked toolbox of interventions, considerable untapped potential exists to foster impactful climate action.

Several limitations should be noted. Our studies focused on two Western countries and thus on Western, Educated, Industrialized, Rich, Democratic (WEIRD) samples[65], even though climate change is a global challenge. Results may not generalize across diverse cultural contexts, and future research should test their robustness internationally with culturally adapted outcome measures. Yet, the USA and Germany are among the top ten CO_2 emitters; thus, climate action is especially necessary there. Furthermore, our research examined only a subset of possible attitudes and behaviours related to climate action which may limit comparability with previous research and generalizability. For instance, we tested actual behaviours that reflect willingness to incur personal costs for climate goals—a key dimension of climate action the Intergovernmental Panel on Climate Change (IPCC)[66] highlighted—but were more often rare than frequent and capture only part of the broader behavioural spectrum from sustained lifestyle changes to collective engagement. Furthermore, whereas willingness was assessed as general willingness, actual behaviour was measured as specific behaviour during our study, which complicates direct comparability between willingness and behaviour. Moreover, it is unclear whether differences in the variance of actual proportions between climate-action types and behaviours were responsible for different degrees of misperceptions between them (Fig. 3) or rather distinct judgement processes, for instance, depending on whether a behaviour can be observed well or not. In addition, lower willingness to donate than in previous work[10] may reflect differences in wording, samples or survey mode. More broadly, because we examined estimated proportions rather than degrees of engagement (for example, mean donations), it remains unclear whether similar misperceptions would arise for more continuous behavioural measures. Therefore, future work should examine the generalizability of our findings to a broader range of attitudes and behaviours, samples and variations in wordings and estimation scales.

In sum, the present findings suggest that pluralistic ignorance in the climate domain is more nuanced than previously assumed and correcting these misperceptions does not seem to be a silver bullet to promote climate action. Instead, efforts may benefit from focusing on more effective avenues for communication and behaviour change. More generally, as in previous analyses of other topics[67], this research also demonstrates how methodological choices shape the understanding of human cognition and behaviour and interventions derived from it. When research on phenomena such as pluralistic ignorance focuses repeatedly on a narrow set of outcome measures, apparent consistency may reflect methodological repetition rather than psychological reality. This can have real-world impact when used for policymaking. Accordingly, behavioural science should remain self-critical and systematically test the generalizability of its findings[68–70], especially if the narrative seems compelling and the evidence points to seemingly easy solutions. Taken together, these results call for greater scrutiny of seemingly consistent findings and long-held assumptions about social perceptions and their role in collective climate action.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41558-026-02668-z.

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Methods

All studies were preregistered and approved by the ethics committee of the University of Erfurt. Preregistrations, original materials, data and code are available at https://osf.io/kcq37 (ref. 33).

Study 1 on donations, policy support and the effect of norm communication (Germany)

Participants. Data were collected in the Planetary Health Action Survey[37] (PACE), a repeated cross-sectional online survey of German citizens aged 18–74 years. Recruitment used quota sampling (age × gender, federal state) via the panel provider Bilendi. In October 2024, 1,130 participants completed the initial survey; 883 (78.1%) participated in a follow-up survey in December 2024. Descriptive statistics are provided in Supplementary Section 3. Power analysis indicated a minimum of 199 participants to detect small effects ($d = 0.20$, $\alpha = 0.05$, $1 - \beta = 0.80$) for comparing estimations and actual proportions for each behaviour or policy. To detect a small effect ($d = 0.20$, $\alpha = 0.05$, $1 - \beta = 0.80$) of norm-communication condition on willingness, at least 788 participants were required.

Procedure. In the initial survey, participants completed the following: (1) support for eight climate policies (7-point scale: 1 = 'strongly disagree' to '7 = strongly agree'); (2) estimates of the proportion of participants somewhat or strongly supporting the policies (0–100%); and (3) a donation task (adapted from Andre et al.[10]). In this task, participants were first asked about their willingness to donate 1% of their household income every month to fight climate change (yes or no; note that we asked about willingness to 'donate', rather than 'contribute' as in Andre et al.[10]). In addition and extending the original task, those who were willing were given the opportunity to donate that amount to the climate action programme of the World Wildlife Fund (WWF, which was widely known and generally rated positively by participants; Supplementary Section 2) by completing a form prefilled with the lower bound of 1% of their income category (assessed at the beginning of the study) on an external webpage. Donations were verified by asking participants to answer questions about information contained in the donation confirmation page or confirmation email, which was only accessible to participants who had actually donated. Then, participants were randomly assigned to one of two conditions. Half of the participants estimated the percentage of others being willing to donate ($n = 565$), while the other half of participants estimated the percentage of others actually donating (0–100%, $n = 565$).

In the follow-up study 8 weeks later, participants completed an adapted WEPT[34,35]. The WEPT is a validated measure to assess actual pro-environmental behaviour by letting participants choose to invest effort and time to trigger actual donations to an environmental organization. Using this task responds to the request of using more validated behavioural tasks in behavioural climate science[36]. To trigger donations, participants are asked to identify numbers belonging to a predefined class within a large array of numbers presented on a page. Here, participants could work through up to six such pages and stop the task whenever they wanted. For each completed page, participants were told that the experimenter would donate a small amount to the WWF if they were among a randomly selected subset of participants. Participants were told the maximum number of possible pages and could choose whether to complete a page or continue with the survey. We considered a page completed if participants identified at least one number. Subsequently, all participants estimated what percentage of participants completed at least four of the six pages. Next, for the norm-communication experiment, participants were randomly assigned to one of two conditions. They were either informed that in the initial survey, 4% of participants had actually donated to the WWF climate-action programme when given the opportunity after indicating willingness and 96% had not (low-prevalence condition, $n = 441$) or received no information (control, $n = 442$). They then reported intentions to discuss climate change with others, sign petitions and donate money to fight climate change in the next year (7-point scale, ranging from 1 = 'never' to 7 = 'always').

Analysis. As preregistered and used in other work on pluralistic ignorance[13], we tested deviations of estimated proportions from actual proportions using one-sample t-tests. This comparison is justified because participants estimated the proportion of people in a specific realized study sample, not in a broader population. The observed proportion therefore serves as the fixed benchmark for the exact reference group being judged, rather than as an estimate of an external population parameter with its own sampling error. Specifically, preregistered analyses included one-sample t-tests comparing estimated proportions to actual proportions (donations, WEPT and support for each individual policy) and a multilevel regression of deviations of estimations from the actual proportions on actual proportions, with random intercepts for participants. To test the effect of norm communication on intentions, three independent-sample t-tests were conducted comparing the low-prevalence and control conditions. All statistical tests reported in this article were two-sided.

Study 2 on participation in a constitutional complaint (Germany)

Participants. In July 2024, 628 participants took part in study 2's initial survey. This study was part of a larger experiment in which only two of the eight conditions were relevant for this research. For study 2, participants from five previous PACE waves (excluding study 1, $n = 5{,}081$) were invited to participate, of which 2,409 completed study 2's initial survey. Of those, 1,998 participants (82.9%) completed the follow-up study 8 weeks later, of which 514 (81.8%) belonged to the two conditions relevant to the current research (willingness condition, $n = 258$; behaviour condition, $n = 256$). Samples of the five PACE waves were each quota-based for age × gender and federal state. Descriptive statistics are provided in Supplementary Section 3. Power analysis indicated that 199 participants were required to detect a small effect ($d = 0.20$, $\alpha = 0.05$, $1 - \beta = 0.80$) for differences between estimated and actual proportions for each outcome (willingness and behaviour), resulting in a total required sample size of 398 participants.

Procedure. The initial survey informed participants about a constitutional complaint against the German government (Action for Future, https://zukunftsklage.greenpeace.de/) and asked about their own willingness to participate as a claimant, including the possibility that they had already participated. All German residents aged 14 years or older could participate as claimants, and participation did not require any financial resources. Then, participants were randomly assigned to one of two conditions. In the willingness condition, participants estimated the percentage of study participants willing to participate. Next, all participants who stated that they were willing could request paperwork to join the constitutional complaint by filling out an online form on an external webpage, after which they would receive paperwork to sign and return to the law firm. Subsequently, all participants estimated the percentage of study participants actually participating in the constitutional complaint. Afterwards, they were also asked about their willingness to donate to support the constitutional complaint. Then, participants in the willingness condition estimated the percentage of study participants willing to donate. At the end of the study, information was provided to all participants on how to donate to support the constitutional complaint.

In the follow-up survey, participants reported whether they had participated and/or donated, verified via answering questions based on the information they had received via email and which only participants who had actually participated and/or donated could answer. Further, all participants were asked to estimate the percentage of people who actually donated.

Analysis. Four preregistered one-sample t-tests compared estimated and actual proportions of willingness and behaviour (each for participation and donations). As preregistered, for actual proportions, we collapsed the data of the willingness and behaviour conditions. When analysing the estimated proportions regarding willingness, we only analysed data from the willingness condition. When analysing the estimated proportions regarding behaviour, we only analysed data from the behaviour condition.

Study 3 on various types of climate action (USA)

Participants. In November 2024, 1,112 US participants (quota-representative for age, gender, education and region) were recruited via the panel provider Bilendi. Descriptive statistics are provided in Supplementary Section 3. To detect a small effect size ($d = 0.20$, $\alpha = 0.05$, $1 - \beta = 0.80$) when comparing estimated and actual proportions, at least 199 participants were required (per comparison). As there were four different questions about the beliefs about other people's donation behaviour ('Procedure' section), at least 796 participants were required. We aimed to recruit 1,100 participants to obtain reliable estimates of the actual proportions.

Procedure. In the study's first part, participants reported whether or how often they performed 20 climate-friendly behaviours (for example, 'I repair items that can still be fixed'; for ten items on a dichotomous yes–no scale; for the other ten items on a 5-point scale ranging from 1 = 'never' to 5 = 'always') and stated the degree to which they supported 20 climate policies (for example, 'Setting stricter fuel efficiency standards for new cars'; 7-point scale, ranging from 1 = 'strongly oppose' to 7 = 'strongly support'). Behaviours and policies were selected to represent diversity in terms of the prevalence of performance and public support according to previous surveys. Next, participants completed a ten-page version of the WEPT (see study 1; here, the donations went to the WWF's climate action programme), and we assessed how many of those ten pages they completed (0–10). Finally, participants were randomly assigned to one of two bonus conditions. In the US$1 condition ($n = 534$), they were given a US$1 bonus. In the US$50 condition ($n = 578$), they could win a US$50 bonus (with the expected value being around US$0.87). Then, all participants could decide to donate any proportion of their bonus (up to US$1 or up to US$50) to atmosfair, a climate-action organization, and we assessed the proportion of the bonus participants donated (0–1).

In the second part, participants provided estimates of others' attitudes and behaviours. First, they estimated the percentage of participants who performed (randomly selected) 8 of the 20 climate-friendly behaviours (that is, chose 'yes' in dichotomous items or 'often' or 'always' in 5-point items). Second, they estimated the percentage of participants who somewhat or strongly supported (randomly selected) 8 of the 20 climate policies (that is, chose 5, 6 or 7 on the 7-point scale). Third, they provided an estimate regarding WEPT behaviour. For this, they were randomly assigned to one of two conditions. To study perceptions of behaviour shown by a large proportion of participants, we asked participants in the at-least-one-page condition to estimate the percentage of participants who completed at least one page (anticipated to be a large proportion; $n = 585$). To study perceptions of behaviour of a small proportion of participants, we asked participants in the all-pages condition to estimate the percentage of participants who completed all ten pages (anticipated to be a small proportion; $n = 527$). The same rationale was applied to the following estimations of donations, for which participants were again randomly assigned to one of two conditions: To study perceptions of majority behaviour, participants in the some-amount conditions estimated the percentage of participants donating at least some amount of their bonus (that is, >US$0 of US$1 or US$50, depending on bonus condition; $n = 563$). To study perceptions of minority behaviour, participants in the whole-amount condition estimated the percentage of participants donating the whole amount ($n = 549$). Finally, participants reported perceived frequency of climate-change news coverage and friends and family bringing up climate change in conversations, with one item for each (7-point scale, ranging from 1 = 'never' to 7 = 'always').

Analysis. To test regression to the mean in climate-friendly behaviours and support for policies, we conducted two preregistered multilevel regressions, with random intercepts for participants. Actual proportion was the predictor variable, and deviation of estimated from actual proportion was the outcome variable. To compare estimated and actual proportions with regard to WEPT behaviour (at least one page or all pages) and donations (at least something or whole amount, for each of the US$1 and US$50 conditions), we conducted preregistered one-sample t-tests.

Computational modelling. We modelled estimated proportions using the Uncertainty-Based Rescaling model of Guay et al.[22], which describes the estimated proportion, $\Psi(P_{\mathrm{actual}})$, as a non-linear transformation of the actual proportion, P_{actual}, as follows (equation (1); see Guay et al.[22] for a derivation):

$$\Psi\left(P_{\mathrm{actual}}\right) = \frac{\delta^{(1-\gamma)} P_{\mathrm{actual}}^{\gamma}}{\delta^{(1-\gamma)} P_{\mathrm{actual}}^{\gamma} + \left(1 - P_{\mathrm{actual}}^{\gamma}\right)}. \tag{1}$$

In this formulation, the parameter δ governs the overall elevation of the estimates, with greater values reflecting generally higher estimates. The parameter γ governs the sensitivity to differences in actual proportions, with larger values (≤1) reflecting greater differentiation.

We estimated the model parameters in a hierarchical Bayesian framework using the brms package[71] (v2.21.0) in R. In our hierarchical extension, we allowed both parameters to vary across individuals by including participant-level random effects on δ and γ. Following Guay et al.[22], we left δ unconstrained, allowing any positive value (bounded below at 0), while constraining γ to the interval [0,1] by estimating an unbounded latent parameter γ_{raw} and mapping it to [0,1] with a logit link $\gamma = \mathrm{logit}^{-1}(\gamma_{\mathrm{raw}})$.

We placed weakly informative priors on the group-level parameters (normal(0.5,1) for δ; normal(0,1) for γ_{raw}) and exponential(1) priors on the standard deviations of participant-level deviations. The model was fit using three Markov chain Monte Carlo chains (MCMC) of 10,000 iterations each (1,000 warm-up). We verified chain convergence via trace plots, effective sample sizes and $\hat{R}$. $\hat{R}$ values were below 1.01 and bulk and tail effective sample sizes exceeded 2,600 and 5,600, respectively, indicating good convergence[72]. Posterior distributions were transformed back to the interpretable scales of δ and γ, and the model fit was evaluated using Bayesian R^2 (ref. 73).

To examine which interindividual differences were associated with δ (elevation) and γ (differentiation), we ran two linear regressions, one for each parameter. The outcome variable was the mean of the individual-level posterior distribution, and the predictor variables were age, gender (dummy coded; 1 = male, 0 = female/other), college degree (dummy coded; 1 = yes, 0 = no), climate-friendly behaviour (sum of 20 behaviours, either shown (that is, response is 'yes', dichotomous items) or shown often or always (frequency items); range: 0–20), policy support (for consistency with the behaviour scale, sum of the policies a person somewhat or strongly supported out of the 20 policies; range: 0–20), number of completed WEPT pages (range: 0–10), proportion of bonus donated to the climate organization (range: 0–1), the extent to which people perceived the news media to cover climate change and the extent to which people perceived friends and family to bring up climate change in conversations (range: both 1–7). All continuous outcome and predictor variables were standardized before running the regressions.

Studies 4a and 4b on the effect of norm communication on donations (Germany)

Participants. Studies 4a and 4b were each part of further PACE survey waves with German participants (age 18–74 years; quota sampled for age × gender and federal state). In May 2025, 1,092 German participants took part in study 4a. In November 2025, 1,119 German participants took part in study 4b. Descriptive statistics are provided in Supplementary Section 3. Power analysis indicated that, per study, at least 964 participants were required to find a small effect ($w = 0.10$, $\alpha = 0.05$, $1 - \beta = 0.80$) when comparing willingness and behaviours across conditions with χ^2 tests.

Procedure. Study 4b was a direct replication of study 4a; thus, both studies were methodologically equivalent. Participants were randomly assigned to one of three conditions. In the low-prevalence condition (study 4a, $n = 354$; study 4b, $n = 362$), they were informed that 4% of participants in a study donated 1% of their household net income monthly to fight climate change (referring to study 1), with a graphic visualizing the proportion (adapted from Andre et al.[9]), and, in smaller font below, that 96% did not. In the high-prevalence condition (study 4a, $n = 373$; study 4b, $n = 379$), they were informed that a study found that 68% were willing to contribute 1% of their household income monthly to fight global warming (study 4a) or climate change (study 4b; referring to the results of the German sample of Andre et al.[10]; this wording was the only difference between studies 4a and 4b besides the sample), with a graphic visualizing the proportion, and, in smaller font below, that 32% were not willing. In the control condition (study 4a, $n = 365$; study 4b, $n = 378$), participants received no information. All participants then reported donation willingness and could donate 1% of their household net income to the WWF climate-action programme, as in study 1.

Analysis. As preregistered, we compared willingness and behaviour across conditions using χ^2 tests and z tests for proportions for pairwise comparisons. In addition, we analysed the pooled data of both studies with two logistic regressions that included two dummy condition variables (reference: control condition) and study (study 4a versus study 4b; dummy coded) as predictor variables and either donation willingness or behaviour as the outcome variable.

Ethics statement

The research was performed in accordance with the general ethical guidelines established by the University of Erfurt and the German Research Foundation. All studies were approved by the Ethics Committee of the University of Erfurt (approval nos. 2024-01, 2024-29 and 2025-25).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All preregistrations, materials and data are publicly available at https://osf.io/kcq37 (ref. 33). Source data are provided with this paper.

Code availability

All custom code developed for this study is publicly available at https://osf.io/kcq37 (ref. 33).

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Author contributions

K.E.T. conceptualized the research and administered the project. K.E.T., K.M. and C.B. designed the studies, developed the methodology and carried out the investigation. K.E.T. led data curation and formal analyses, with support from K.M. K.E.T. created the visualizations and drafted the original manuscript. C.B. acquired funding. All authors contributed to writing, reviewing and editing the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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Software and code

Policy information about availability of computer code

Data collection	For all studies, data were collected using the online survey software Unipark. Participants were recruited via the panel provider Bilendi.
Data analysis	For all studies, data were analyzed using R (v4.4.1) and RStudio (v2025.05.0). Custom analysis scripts were implemented using established R packages, including brms (v2.21.0) for hierarchical Bayesian modeling, lme4 (v1.1-35.5) and lmerTest (v3.1-3) for mixed-effects modeling, and effectsize (v0.8.9) for effect size estimation. Data processing and visualization were conducted using the tidyverse (v2.0.0), including ggplot2 (v4.0.0). All custom analysis scripts are available at https://osf.io/kcq37.

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Reporting on sex and gender	Gender was assessed via self-report. Participants were asked to indicate their gender identity using predefined response options (woman, man, diverse/non-binary/other, in one study additionally "prefer not to say"). Biological sex was not assessed. Gender was used descriptively to characterize the samples and analytically to examine associations between gender and estimation deviations. Gender was also used to define quota targets during participant recruitment. No analyses were restricted to a single gender, and the studies were not designed to test gender-specific effects.
Reporting on race, ethnicity, or other socially relevant groupings	Race and ethnicity were not assessed in any of the studies and were therefore not included in the analyses.
Population characteristics	see below
Recruitment	For all studies, participants were recruited via the panel provider Bilendi. Participants were informed that the studies concerned climate change, which may have introduced self-selection bias (e.g., overrepresentation of individuals with higher interest in or engagement with climate issues). This could lead to higher baseline levels of pro-climate attitudes and behaviors, potentially inflating absolute estimates. However, because the main analyses focused on differences between estimated and actual proportions and on experimental effects, such bias is unlikely to fully account for the observed patterns.
Ethics oversight	All studies were approved by the ethics committee of the University of Erfurt.

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Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This research comprised five preregistered online experiments examining perceptions of others' climate-related attitudes, intentions, and behaviors. Across studies, participants reported their own climate-related attitudes or behaviors and estimated the prevalence of these outcomes among other participants. Some studies additionally measured verified behavior (e.g., donations, political participation, and effort-based tasks) and experimentally manipulated norm information to test the effects of communicating high versus low prevalence of climate action. The studies combined descriptive comparisons of estimated and actual proportions with experimental and computational modeling approaches to examine the generalizability, mechanisms, and behavioral consequences of pluralistic ignorance.
Research sample	The research samples consisted of adult participants (aged 18–74 years) from Germany and the United States recruited via online panel providers (Bilendi). Quota-based sampling was used to approximate population distributions with respect to age, gender, education (US sample only), and region/federal state. Thus, the samples were approximately representative of the population with respect to these characteristics, but not fully representative of the general population. Detailed demographic characteristics are reported in the manuscript and Supplementary Information. The use of large online panel samples allowed for efficient recruitment of diverse participants and sufficient statistical power to detect small effects across multiple experimental studies. Germany and the United States were selected as high-income, high-emission countries in which climate action is particularly relevant, and where quota-based online panels enable approximation of population distributions.
Sampling strategy	Sample sizes were determined a priori based on power analyses. Participants were recruited via the online panel provider Bilendi using quota-based sampling. Full details are provided in the Methods section of the manuscript.
Data collection	Data were collected entirely online using the survey software Unipark. Participants completed all questionnaires, behavioral tasks, and donation decisions remotely and independently on their own devices. No researcher or other participants were present during data collection; all interactions were fully automated via the survey platform. Because data collection and experimental manipulations were implemented programmatically within the survey software, researchers were not directly involved in assigning participants to conditions. Thus, researcher blinding to experimental condition was not applicable. Similarly, participants were not informed about the specific study hypotheses to minimize potential demand characteristics.

Timing	Data were collected across multiple online survey waves between 2024 and 2025. Detailed information on the timing of the individual studies is provided in the manuscript.
Data exclusions	Data exclusions were applied only as preregistered. Across all studies, analyses included only participants who completed the questionnaires. In Study 3, participants were additionally to be excluded if their completion time fell below a preregistered minimum duration threshold; however, no participants met this criterion. Thus, no participants were excluded from the analyses in any of the studies.
Non-participation	Attrition occurred in studies involving follow-up surveys. In Study 1, 883 of 1,130 participants (78.1%) completed the follow-up survey. In Study 2, 1,998 of 2,409 participants (82.9%) completed the follow-up survey. No information on reasons for non-completion was collected. Analyses were conducted on participants who completed the respective study components.
Randomization	Participants were randomly allocated to experimental conditions using the built-in randomization procedure of the survey software Unipark. Allocation was automated and therefore not influenced by participant characteristics or researcher involvement.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*